

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOBIDIR

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'rinishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!
28.02.2026. 2-smena.

1. Quyidagi sonni 7 ga bo'lgandagi qoldiqni toping:

$$\underbrace{77 \dots 79}_{2025 \text{ ta}} + \underbrace{77 \dots 78}_{2025 \text{ ta}} + \dots + \underbrace{77 \dots 71}_{2025 \text{ ta}} .$$

- A) 1
- B) 3
- C) 5
- D) 4

2. Hisoblang:

$$4\frac{2}{2025} \cdot 6\frac{5}{2026} - 2\frac{2023}{2025} \cdot 7\frac{2021}{2026} - \frac{35}{2026} .$$

- A) $\frac{28}{2025}$
- B) 1
- C) 0
- D) $\frac{35}{2026}$

3. Ikki shahardan bir-biriga qarab ikk jism yo'lga chiqdi birinchisi 1 soat o'tganda 2-dan 2 km ko'p yo'l bosib o'tdi. Agar birinchi jism ikkinchisidan 1 soat oldin manzilga yetib kelgan bo'lsa ikkinchi jism qancha vaqtda yetib kelgan? Ikki shahar orasi 24km.

- A) 3
- B) 5
- C) 6
- D) 4

4. 2 ta bir xil narxdagi kurtka. Sotuvchi ulardan birini 5 foiz chegirma bilan sotdi. Ikkinchisini chegirmasiz. Agar 1-sidan olingan foyda kurtka 18% ini tashkil qilsa, 2-chi kurtkadan olingan foyda uning o'zining narxining necha foizini tashkil qiladi?

- A) 21
- B) 22
- C) 23
- D) 17

5. Hisoblang:

$$\frac{(3^4 \cdot 2)^{-\frac{1}{2}}}{9^{-2} \cdot 2^{-2}} .$$

- A) 18
- B) $18\sqrt{2}$
- C) $9\sqrt{2}$
- D) 9

6. Hisoblang:

$$\sqrt[6]{7 - \sqrt{40}} \cdot \sqrt[3]{\sqrt{2} + \sqrt{5}} \cdot \sqrt[3]{9}.$$

- A) 2
- B) 3
- C) 4
- D) $3\sqrt{2}$

7. Hisoblang:

$$\frac{5}{\sqrt[3]{32} + 2} + \frac{5}{\sqrt[3]{128} - \sqrt[3]{32} + 2} - \sqrt[3]{2}.$$

- A) 2
- B) 0
- C) $\sqrt[3]{2} + 1$
- D) 1

8. $a_n = 3,8 - 0,7n$ qonuniyat bilan aniqlangan ketma-ketlik berilgan. Shunga ko'ra S_{11} ni toping.

- A) $-4,4$
- B) -44
- C) $-5,4$
- D) $4,4$

9. Agar kamayuvchi geometrik progressiyada $b_1 + b_2 + b_3 = 19$ va $b_1^2 + b_2^2 + b_3^2 = 133$ bo'lsa, uning dastlabki 5 ta hadi yig'indisini toping.

- A) $23\frac{2}{9}$
- B) $23\frac{1}{9}$
- C) $22\frac{7}{9}$
- D) $23\frac{4}{9}$

10. $a = \sqrt{2}$ va $b = \sqrt{18}$ bo'lsa, quyidagini soddalashtiring:

$$\left(\frac{\sqrt{a}}{\sqrt{a} - \sqrt{b}} - \frac{\sqrt{b}}{\sqrt{a} + \sqrt{b}} - \frac{2\sqrt{ab}}{a - b} \right) \cdot \left(\frac{\sqrt{3}a}{a - \sqrt{ab}} + \frac{\sqrt{3}b}{\sqrt{ab} - b} \right).$$

- A) 3
- B) $\sqrt{3}$
- C) 1
- D) $\sqrt{6}$

11. Soddalashtiring:

$$(2x - 1)^4 - 8x(2x - 1)^3 + 24x^2(2x - 1)^2 - 32x^3(2x - 1) + 16x^4 .$$

- A) 0
- B) $(4x - 1)^2$
- C) 1
- D) -1

12. Tenglamaning $\left[\frac{\pi}{2}; \frac{5\pi}{2}\right]$ oralig'ida nechta ildizi bor:

$$\cos\left(\frac{3\pi}{2} + 2x\right) = 0 .$$

- A) 3 ta
- B) 5 ta
- C) 4 ta
- D) 6 ta

13. Tenglamalar sistemasini yechimlari uchun $x \cdot y$ ni toping:

$$\begin{cases} x^{3y-4} = 9 \\ x^{y+1} = 27 . \end{cases}$$

- A) 3
- B) 9
- C) 4
- D) 6

14. Tenglamaning haqiqiy ildizlari sonini toping:

$$\frac{\sqrt{4-x} - \sqrt{5x^3 - 4x^2 - 4x}}{\sqrt{4-x} + \log_{(5x+1)}^2(5x^3 - 4x^2 + 1)} = 1 .$$

- A) 2 ta
- B) 3 ta
- C) Haqiqiy ildizga ega emas
- D) 1 ta

15. Tenglamaning haqiqiy ildizlari sonini toping:

$$x^4 - \frac{2027!}{2026! + 2025!} \cdot x^2 + 2025 = 0 .$$

- A) 3 ta
- B) 2 ta
- C) 5 ta
- D) 4 ta

16. Tenglamani yeching:

$$x + \frac{x}{1+2} + \frac{x}{1+2+3} + \cdots + \frac{x}{1+2+\cdots+4041} = 4041 .$$

- A) 2022
- B) 2021
- C) 2020
- D) 4041

17. Tengsizlikning butun yechimlari sonini toping:

$$\sqrt{x^2 - 4x + 4} < 5 \cdot \sqrt{2 - x} + 6 .$$

- A) 33 ta
- B) 32 ta
- C) 34 ta
- D) 36 ta

18. Tengsizlikning butun yechimlari yig'indisini toping:

$$\frac{|x^2 - 8x + 12|}{18 + 7x - x^2} > 0 .$$

- A) 35
- B) 27
- C) 36
- D) 19

19. Tenglama $\left[-\frac{3\pi}{2}, \frac{5\pi}{2}\right]$ kesmada nechta ildizga ega?

$$\sin^4 x + \cos^4 x = 1 .$$

- A) 9 ta
- B) 10 ta
- C) 8 ta
- D) 7 ta

20. Funksiyaning aniqlanish sohasini toping:

$$y = \log_{x+2} \frac{(x-2)\sqrt{24+2x-x^2}}{x-2} .$$

- A) $(-2; 0) \cup (0; 2) \cup (2; 6)$
- B) $(-2; -1) \cup (-1; 2) \cup (2; 6)$
- C) $(-2; 2) \cup (2; 6)$
- D) $(-1; 2) \cup (2; 6)$

21. $f(x)$ funksiya uchun $f(x) = \frac{x^4 + 1}{x^2 + 1}$ o'rinli bo'lsa, funksiyaning qiymatlar sohasini toping.

- A) $[2\sqrt{2}; \infty)$
- B) $[2\sqrt{2} + 2; \infty)$
- C) $[2\sqrt{2} - 2; \infty)$
- D) $(2\sqrt{2} - 2; \infty)$

22. $f(x) = (x - 8)e^{x-7}$ funksiyaning $[5; 9]$ oraliqdagi eng katta qiymatini toping.

- A) e^3
- B) e
- C) e^{-2}
- D) e^2

23. Aniq integralni hisoblang:

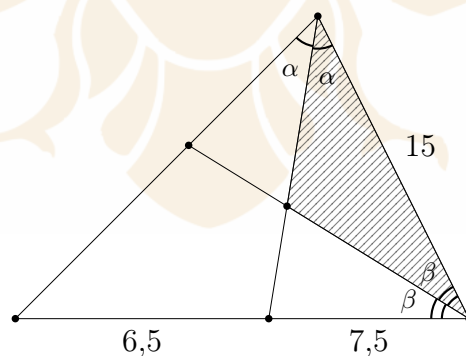
$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin^3 x}{\cos^7 x} dx .$$

- A) 0
- B) 1
- C) $\sqrt{3}$
- D) $2\sqrt{3}$

24. Aylananing 8 ga teng vatari 90° li yoyni tortib turadi. Aylana radiusini toping.

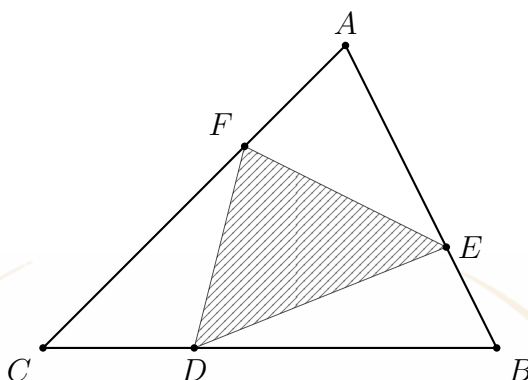
- A) 8
- B) 4
- C) $8\sqrt{2}$
- D) $4\sqrt{2}$

25. Quyidagi rasimga asosan bo'yalgan soha yuzini toping.

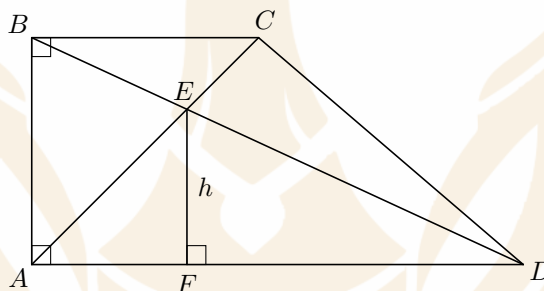


- A) $15\sqrt{3}$
- B) $14\sqrt{3}$
- C) 28
- D) 30

26. ABC va FED muntazam uchburchaklar rasmdagidek joylashtirilgan. $AB = 18$ va $AF : FA = CD : DB = BE : EA = 1 : 2$ bo'lsa, S_{DEF} ni toping.



- A) $27\sqrt{3}$
 B) $18\sqrt{3}$
 C) 27
 D) $9\sqrt{3}$
27. To'g'ri burchakli trapetsiya asoslari 9 va 3 ga, kichik yon tomoni 6 ga teng. Trapetsiya diagonallari kesishgan nuqtadan katta asosgacha bo'lgan eng qisqa masofani toping.



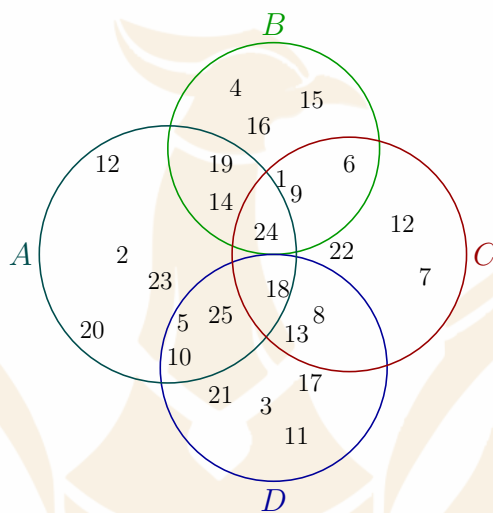
- A) 4
 B) 5
 C) 3,5
 D) 4,5
28. Muntazam ko'pburchakning diagonallari soni 35 ta bo'lsa, bir uchidan chiqqa eng katta va eng kichik diagonallari orasidagi burchak.
- A) 72°
 B) 54°
 C) 36°
 D) 45°
29. $A(3; 1; 2)$, $B(-1; 1; 4)$ va $C(0; 2; 2)$ uchburchakning uchlari bo'lsa, C uchidan AB tomonga tushirilgan mediana uzunligini toping.
- A) $\sqrt{6}$
 B) $\sqrt{10}$

- C) $\sqrt{20}$
D) $\sqrt{3}$

30. Muntazam oltiburchakli piramidaning yon qirrasasi asos tekisligi bilan 60° hosil qiladi. Agar qirrasining uzunligi $\sqrt{39}$ bo'lsa, uning hajmini toping.

- A) $\frac{117\sqrt{39}}{16}$
B) $\frac{117\sqrt{39}}{32}$
C) $\frac{117\sqrt{39}}{8}$
D) $\frac{117\sqrt{39}}{4}$

31. $(A \cup C) \cap (B \cup D)$ ning elementlari sonini toping.

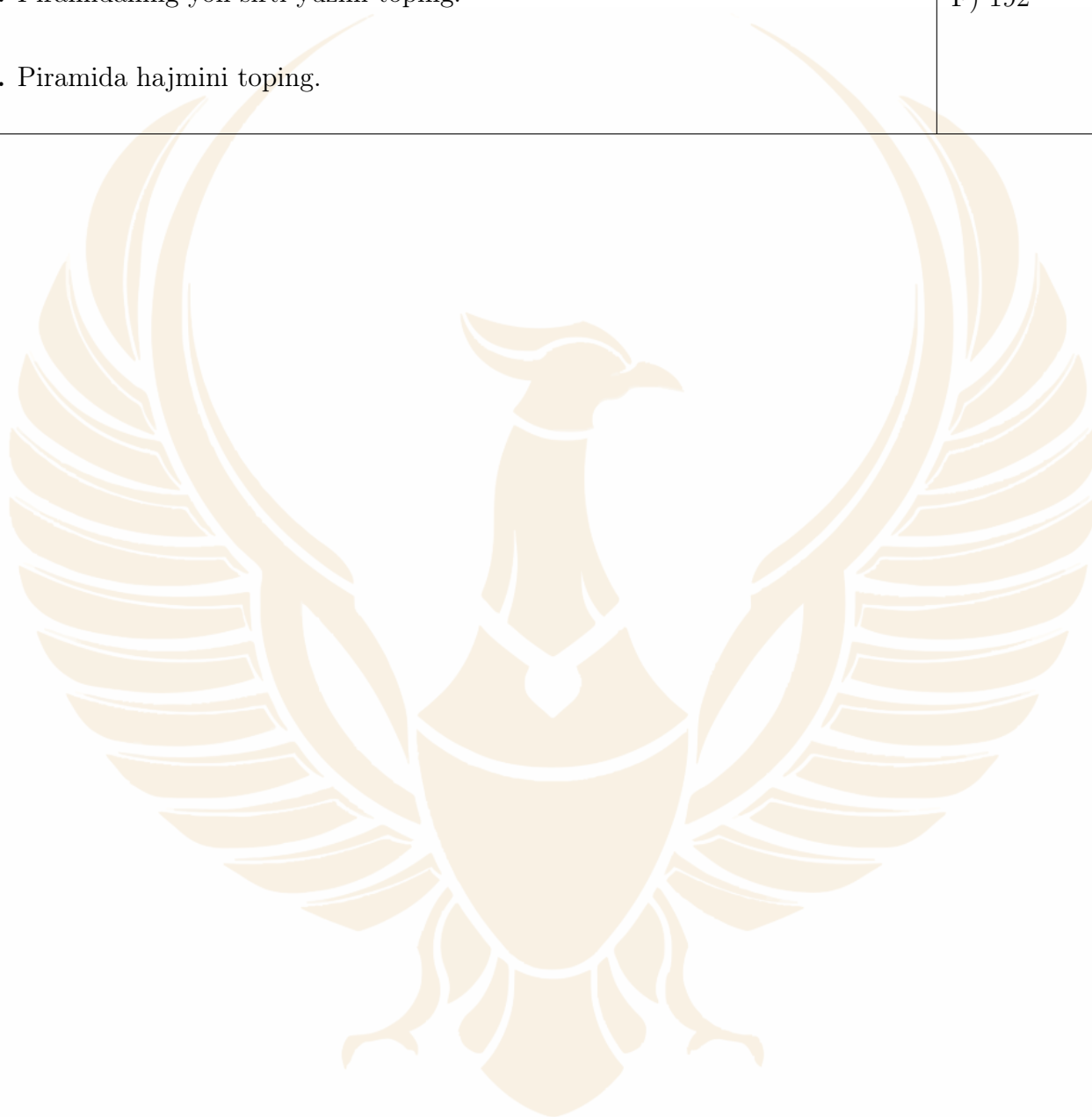


- A) 12
B) 11
C) 13
D) 16

32. “MATEMATIKA” so‘zidagi harflar qog‘ozga alohida yozib chiqilgan. Harflar ichidan tasodifiy tanlanganda M harfi chiqish ehtimolini toping.

- A) $\frac{1}{10}$
B) $\frac{1}{5}$
C) $\frac{1}{4}$
D) $\frac{2}{5}$

Topshiriqlar (33–35) va javob variant ($A–F$) larni o‘zaro moslashtiring.	Javoblar.
Muntazam uchburchakli piramidaning yon yoqlari asos tekisligi bilan 60° burchak hosil qiladi. Piramidaga ichki chizilgan konusning asosi raduisi 4 ga teng.	A) 48
33. Konusning yon sirti yuzini toping.	B) 96
34. Piramidaning yon sirti yuzini toping.	C) $64\sqrt{3}$
35. Piramida hajmini toping.	D) $96\sqrt{3}$
	E) 64
	F) 192



36. Quyidagi tenglamalar sistemasi berilgan:

$$\begin{cases} x - \frac{1}{4}xy - y = 0 \\ x^2 - \frac{5}{2}xy + y^2 = 0 \end{cases}.$$

a) Tenglamalar sistemasining yechimlari uchun $x \cdot y$ ning eng katta qiymatini toping.

Javob: _____

b) Tenglamalar sistemasining yechimlari juftligi nechta?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiiga ko'chirib yozing.

37. Quyidagi tenglama berilgan:

$$\sin^2 x + \sin 2x = 3 \cos^2 x .$$

a) Tenglamaning eng katta manfiy yechimini toping.

Javob: _____

b) Tenglamaning $[-\pi; \pi]$ oraliqdagi yechimlari sonini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiiga ko'chirib yozing.

38. $f(x)$ va $g(x)$ funksiyalar uchun $f(3x - 2) + 7 \cdot g(x - 5) = x + 1$ va $f(x + 1) + g\left(\frac{x}{3} - 4\right) = 3x$ bo'lsa,

a) $(f(1))^{-1}$ ni toping.

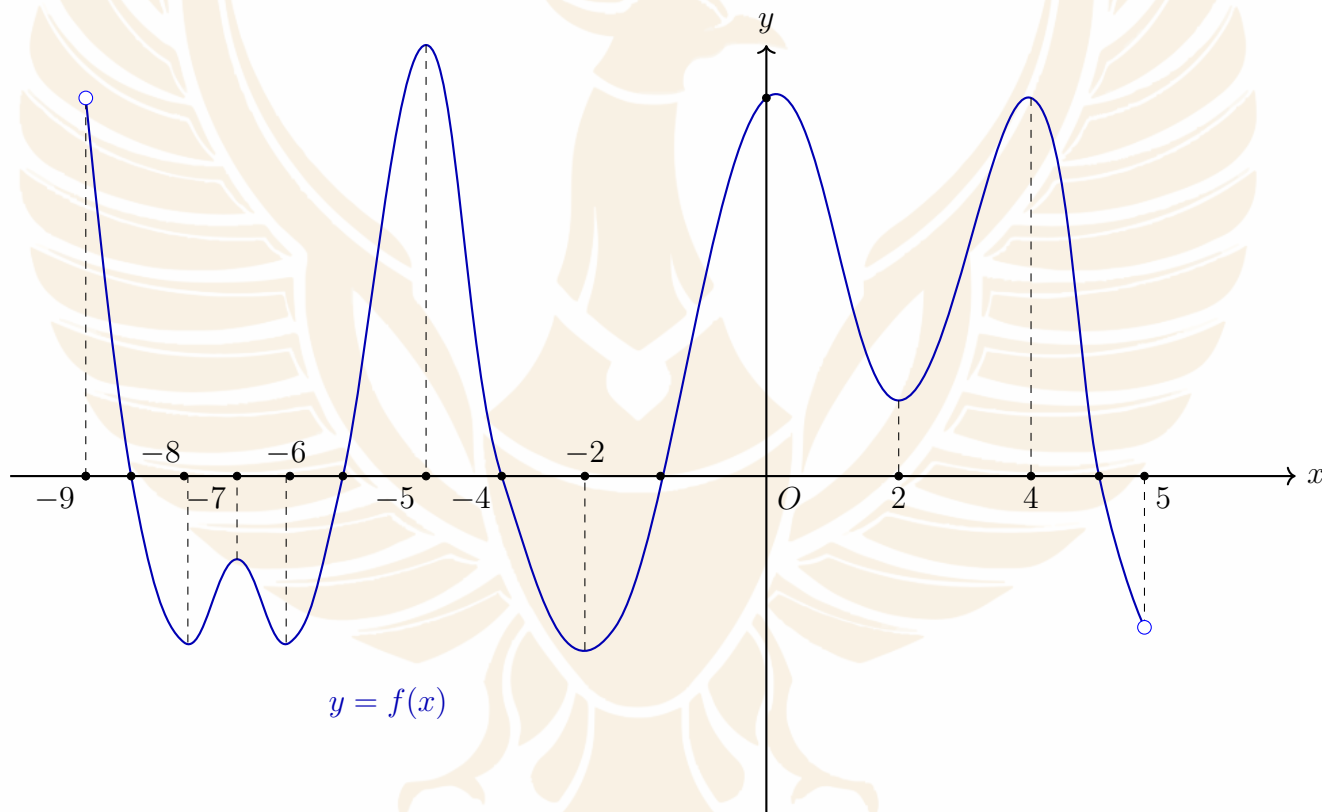
Javob: _____

b) $f(x) + g(x) = \frac{1}{2}$ tenglamaning ildizlari sonini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

39. Quyida $(-9; 5)$ oralig'ida aniqlangan $f(x)$ funksiyaning grafigi tasvirlangan.



a) $f(x)$ funksiyaning ekstremum nuqtalari nechta?

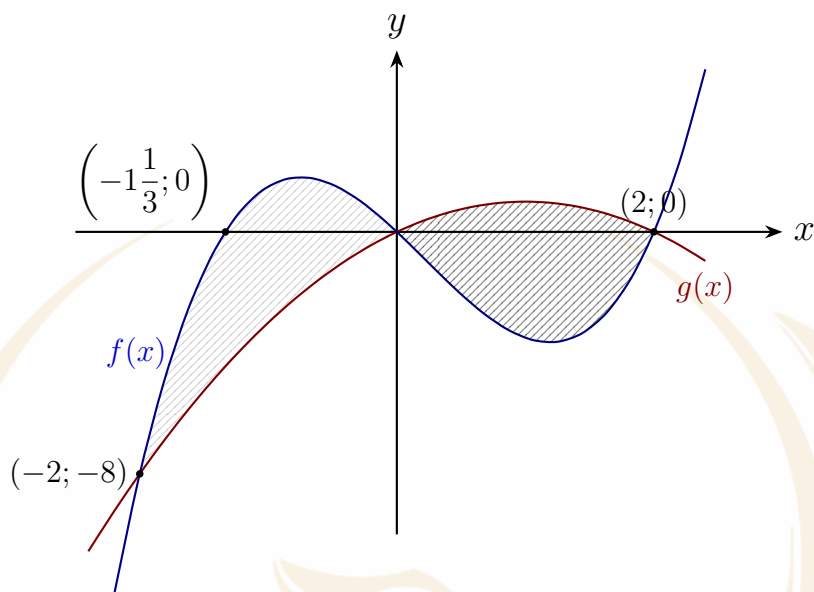
Javob: _____

b) $f'(x) < 0$ tengsizlikni $(-9; 5)$ oralig'idagi nechta butun son qanoatlantiradi.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

40. $f(x) = ax^3 + bx^2 + cx$ va $g(x) = px^2 + qx$ funksiyalar berilgan. Bu funksiyalar $(-2; -8)$, $(0; 0)$ va $(2; 0)$ nuqtalarda kesishadi. $f(x)$ funksiya $\left(-1\frac{1}{3}; 0\right)$ nuqtadan o'tadi.



- a) $\frac{a+b+c}{p+q}$ ni toping.

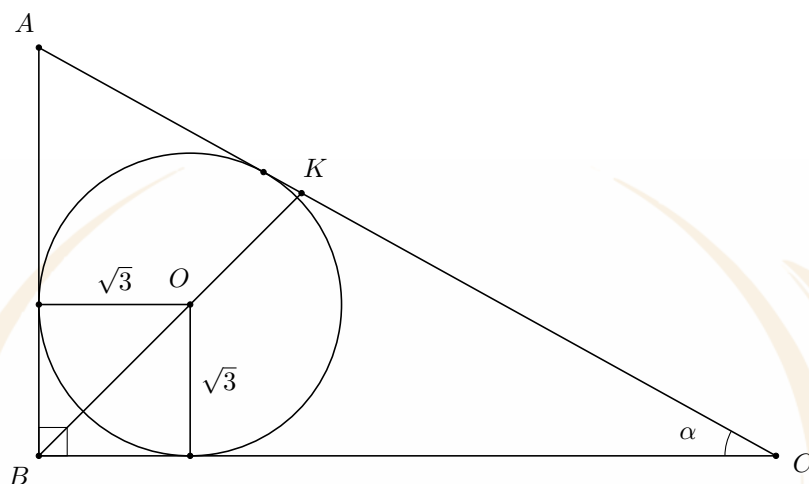
Javob: _____

- b) Bo'yalgan soha yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiqa ko'chirib yozing.

41. ABC to'g'ri burchakli uchburchakga ichki aylana chizilgan. BK bissektisa, $r = \sqrt{3}$ ga teng. $\frac{BO}{OK} = \frac{\sqrt{3}}{\sqrt{2}}$ bo'lsa,



a) OK ni toping.

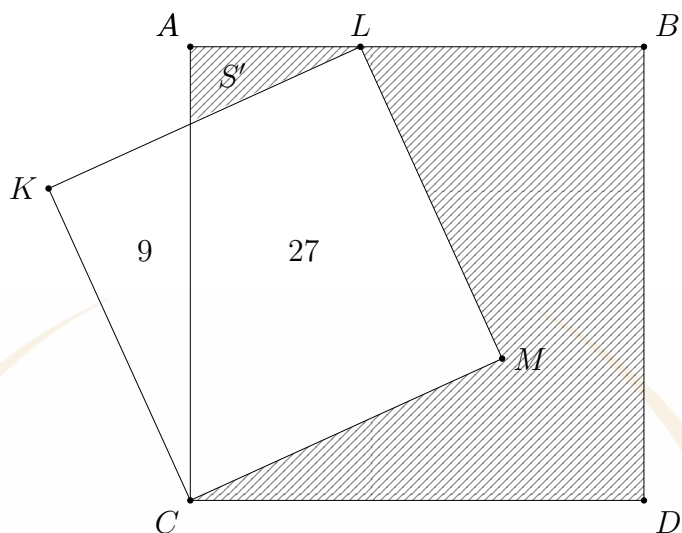
Javob: _____

b) $\angle ACB$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ga ko'chirib yozing.

42. Rasmda $ABDC$ va $KCML$ kvadratlar berilgan.



a) S' yuzani toping.

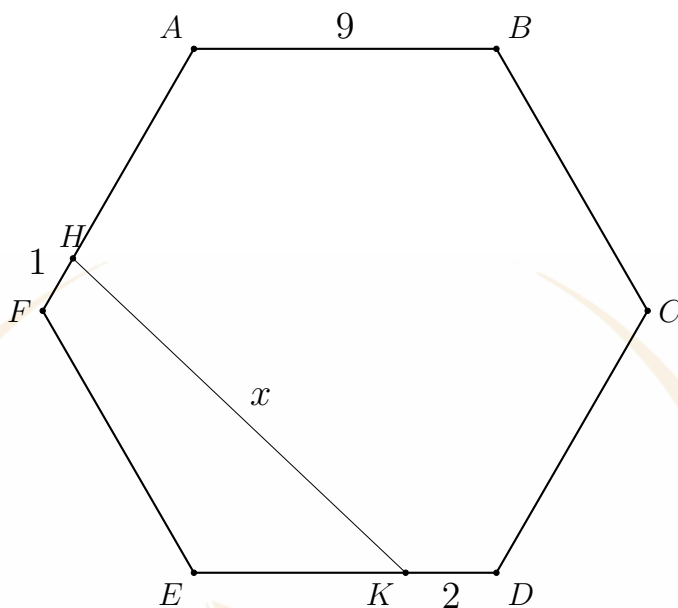
Javob: _____

b) S_{LMCDB} beshburchak yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

43. Muntazam $ABCDEF$ oltiburchakning tomoni uzunligi 9 ga teng. ED tomonidan K va AF tomondan H nuqtalar olinib tutashtirilgan. $FH = 1$, $KD = 2$ va $HK = x$ ga teng.



a) x ni toping.

Javob: _____

b) $\frac{S_{FHKE}}{S_{EBCDKH}}$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqarajasiga ko'chirib yozing.

44. $ABCD$ uchburchakli piramidaga radiusi $\frac{3}{2}$ ga teng bo'lgan shar ichki chizilgan. Shar piramidaning ABD yog'iga va DC qirrasiga hamda $\triangle ABC$ uchburchakning medianalar kesishish nuqtasiga urinadi. $DC = 3$ va $AD \perp AC$ bo'lsa,

a) ADB yoqning AB asosiga tushgan medianasi uzunligini toping.

Javob: _____

b) Piramida hajmini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

45. Bir korxonaning daromad funksiyasi $A(x) = \frac{2x}{3}$ ga, xarajat funksiyasi $B(x) = 2\sqrt{x}$ ga teng. x ishchilar soni.

a) Korxona zarar qilmasligi uchun eng kamida nechta ishchi olishi kerak?

Javob: _____

b) Korxona foyda olishi uchun eng kamida nechta ishchi olishi kerak?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.